

# Taylor Pentz

Senior Rust / Axum / Actix Backend Engineer

High Point, NC

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## Professional Summary

Senior Rust / Axum / Actix Backend Engineer with 11+ years in software engineering across SaaS, fintech, healthcare, analytics, and enterprise workflow products and recent delivery using Rust, Axum, Actix, Tokio, cloud services, databases, automated testing, and CI/CD; modernizes legacy systems, fixes production reliability issues, improves performance, mentors engineers, and turns complex requirements into secure, scalable products with measurable business impact.

## Skills

- **Primary Stack:** Rust, Axum, Actix, Tokio, PostgreSQL, REST APIs; practical delivery across SaaS, fintech, healthcare, analytics, and enterprise workflow products.
- **Backend Depth:** REST, GraphQL, OpenAPI, OAuth, JWT, RBAC, workers, webhooks, idempotency, rate limiting, API gateways, and service observability.
- **Architecture:** microservices, event-driven architecture, domain-driven design, transactional outbox, idempotency; API boundaries, domain modeling, secure design, scalability reviews, and production-readiness planning.
- **Cloud and DevOps:** AWS ECS, Lambda, RDS, SQS, Azure App Service, Docker, Terraform; Docker, CI/CD pipelines, release automation, feature flags, rollback plans, and environment governance.
- **Databases and Messaging:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, SQL Server; schema design, indexing, query tuning, migrations, caching, reconciliation, and reliable async processing.
- **Testing and Quality:** Jest, Supertest, JUnit, Mockito, Pact, integration and contract testing; unit, integration, contract, end-to-end, regression, load, and smoke testing aligned with release risk.
- **Monitoring and Security:** OpenTelemetry, Datadog, CloudWatch, ELK, Sentry, structured logs, alerts, OWASP, OAuth, RBAC, incident response, and RCA.
- **Senior Practices:** technical leadership, code reviews, architecture reviews, mentoring, Agile planning, stakeholder communication, legacy modernization, and production support.

## Work Experience

### SumatoSoft

Senior Software Engineer

Boston, MA Remote

Apr 2023 - Dec 2025

- Led backend architecture for **multi-tenant workflow services** using **Rust, Axum, Actix, Tokio, PostgreSQL, REST APIs**, centered on **Rust 1.70+**, **Tokio**, and **PostgreSQL 14+**, designing service boundaries, API contracts, workers, authorization rules, and database ownership for scalable customer operations.
- Modernized legacy modules with duplicated validation, inconsistent state handling, and fragile release paths into **typed, testable, reusable components and services**, reducing production defects by **32%** across high-change customer workflows.
- Resolved API latency, queue backlog, and duplicate-write issues by adding **index tuning, connection pooling, idempotent commands, async workers, circuit breakers, and payload normalization**.
- Designed secure integration boundaries with **OAuth, RBAC, tenant-aware filters, OpenAPI contracts, audit logging, and defensive validation**, preventing cross-account data exposure during partner and administrator workflows.
- Implemented reliable asynchronous processing with **idempotency keys, retry policies, dead-letter handling, reconciliation jobs, and transactional persistence**, eliminating duplicate business actions after provider timeouts.
- Moved fragile deployments toward **Docker-based CI/CD, infrastructure checks, feature flags, environment promotion controls, and repeatable rollback procedures**, improving release confidence for distributed teams.
- Added automated quality gates with **unit, integration, contract, end-to-end, smoke, and regression tests**, catching broken API contracts, UI edge cases, and migration defects before production deployment.
- Built observability with **OpenTelemetry, structured logs, Datadog and CloudWatch alerts, traces, dashboards, and incident runbooks**, accelerating diagnosis of latency spikes, memory growth, failed jobs, and webhook failures.
- Mentored engineers through **architecture reviews, pair debugging, pull-request standards, incident retrospectives, and performance-focused refactoring**, raising technical consistency across frontend, backend, QA, DevOps, and product groups.

### Hexaview Technologies Inc.

Software Engineer

New York, NY

Jul 2018 - Mar 2023

- Delivered client-facing fintech and analytics features using **Rust for selected services, Node.js APIs, PostgreSQL, Redis, Docker, and Linux systems**, centered on **Rust 1.x for selected services and Node.js 12/14 APIs**, with **OAuth, JWT, RBAC, OpenAPI, Kafka, RabbitMQ, Redis, Docker**, supporting customer dashboards, reporting workflows, account servicing, and external data integrations.
- Refactored older screens, services, and shared libraries that mixed business rules with presentation logic, improving maintainability and reducing regression bug leakage by **29%** across three major release cycles.
- Built service and API integrations for payments, CRM, identity, notifications, and reporting providers, handling **duplicate callbacks, schema drift, partial outages, retries, and reconciliation** without losing customer activity.
- Fixed production memory and worker stability issues by profiling retained objects, closing database cursors, limiting batch size, and introducing safer queue-processing controls for long-running jobs.
- Optimized reporting and search workloads through **composite indexes, query rewrites, cached summaries, asynchronous generation, and defensive timeouts**, stabilizing response times as customer records grew.
- Migrated older modules toward modern runtime, dependency, and build standards while preserving backward-compatible API contracts for dependent clients and scheduled enterprise releases.
- Expanded automated validation with **component tests, API tests, contract fixtures, browser tests, and release smoke suites**, reducing manual QA bottlenecks and catching provider response changes earlier.
- Partnered with architects, product managers, QA, and client stakeholders to split large modernization initiatives into smaller releases with measurable acceptance criteria, release notes, and support documentation.

### SimbirSoft Company

Software Developer

Boston, MA

Mar 2015 - Jun 2018

- Built enterprise workflow, e-commerce, and internal operations modules using **C++, Java, Rust for small service utilities, REST APIs, PostgreSQL, and Linux tooling** with **SQL databases, REST APIs, queues, and CI builds**, supporting approvals, orders, notifications, imports, and reporting.
- Replaced brittle server-rendered and script-heavy flows with cleaner application modules, improving task completion speed by **24%** for operations users working through repetitive approval and data-entry processes.
- Implemented service-layer validation, transaction boundaries, and database constraints that corrected duplicate submissions, inconsistent statuses, and race conditions caused by browser retries or concurrent updates.
- Assisted modernization from manual deployments and tightly coupled code into **repeatable builds, environment profiles, automated tests, structured logging, and safer release packages** across staging and production.
- Improved slow database-backed screens through **index reviews, query tuning, pagination, selective caching, and explain-plan analysis**, reducing support complaints tied to large customer and catalog tables.
- Created REST endpoints and background jobs for notifications, file imports, invoice synchronization, and status tracking while adding error queues and replay scripts for failed integrations.
- Worked with QA, analysts, and senior engineers to clarify requirements, investigate production defects, write integration tests, and document root causes for recurring customer-impacting issues.

## Microsoft

Cambridge, MA

Software Developer Intern

Aug 2014 - Feb 2015

- Contributed to internal developer productivity and reporting tools using **C#, ASP.NET MVC/Web API, JavaScript, TypeScript, SQL Server, and internal Azure-hosted tools**, supporting SQL-backed services, administrative dashboards, validation rules, data exports, and workflow automation for engineering teams.
- Implemented searchable grids, export utilities, form validation, and role-aware visibility rules that helped internal users review operational data and complete routine support tasks more efficiently.
- Investigated defects in data access, validation, and workflow-state handling by reproducing failures, comparing expected outcomes, and preparing fixes reviewed by senior engineers before merge.
- Added unit and integration tests around controller logic, utility classes, SQL queries, and edge-case validation, strengthening regression coverage for tools that had relied heavily on manual checking.
- Participated in code reviews, daily standups, release readiness checks, and design discussions, gaining production-minded engineering habits around maintainability, reliability, and secure implementation.
- Documented setup steps, API assumptions, troubleshooting notes, and known limitations so future interns and full-time engineers could onboard faster and maintain the internal tools confidently.

## Selected Projects

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### Secure API and Workflow Automation Platform

- Designed secure backend workflows with **Rust, Axum, Actix, Tokio, PostgreSQL, REST APIs**, OpenAPI contracts, RBAC, tenant-aware filtering, PostgreSQL models, Redis caching, and reliable integration jobs for customer operations.

### Event-Driven Backend Modernization

- Modernized event-driven services with **idempotency, outbox persistence, retries, dead-letter queues, structured logging, and observability**, reducing duplicate business events and improving support traceability.

## Certifications

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- AWS Certified Developer - Associate
- AWS Certified Solutions Architect - Associate
- Microsoft Certified: Azure Developer Associate
- Certified Kubernetes Application Developer (CKAD)
- Professional Scrum Developer I (PSD I)

## Education

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University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science, 2010 - 2014